

Advanced AQI Analysis Report – Delhi (January 2023)

1. Introduction

This report presents an advanced analysis of hourly air pollution data for January 2023. The objective is to examine pollutant trends, hourly variation, and inter-pollutant relationships.

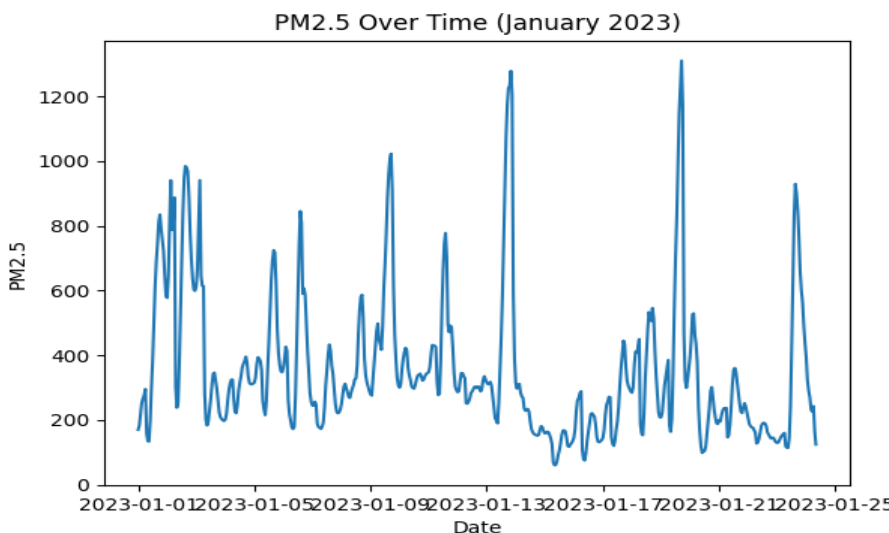
2. Data Loading Code

```
import pandas as pd
df = pd.read_csv("delhiaqi.csv")
df['date'] = pd.to_datetime(df['date'])
df = df.sort_values('date')
df['hour'] = df['date'].dt.hour
```

3. Descriptive Statistics

Average PM2.5: 358.26 $\mu\text{g}/\text{m}^3$ | Maximum PM2.5: 1310.2 $\mu\text{g}/\text{m}^3$ | Average PM10: 420.99 $\mu\text{g}/\text{m}^3$

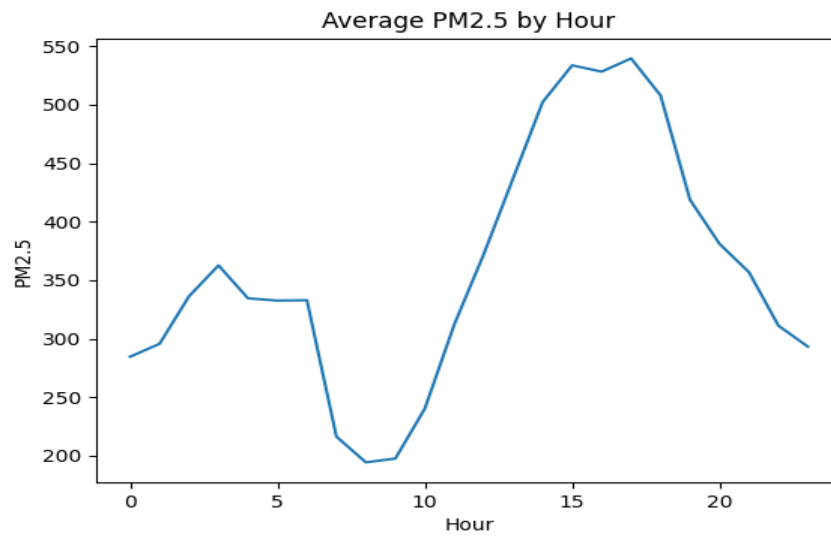
4. PM2.5 Time Trend



PM2.5 levels remain consistently high throughout January, indicating severe winter pollution conditions.

5. Hourly Pattern Analysis

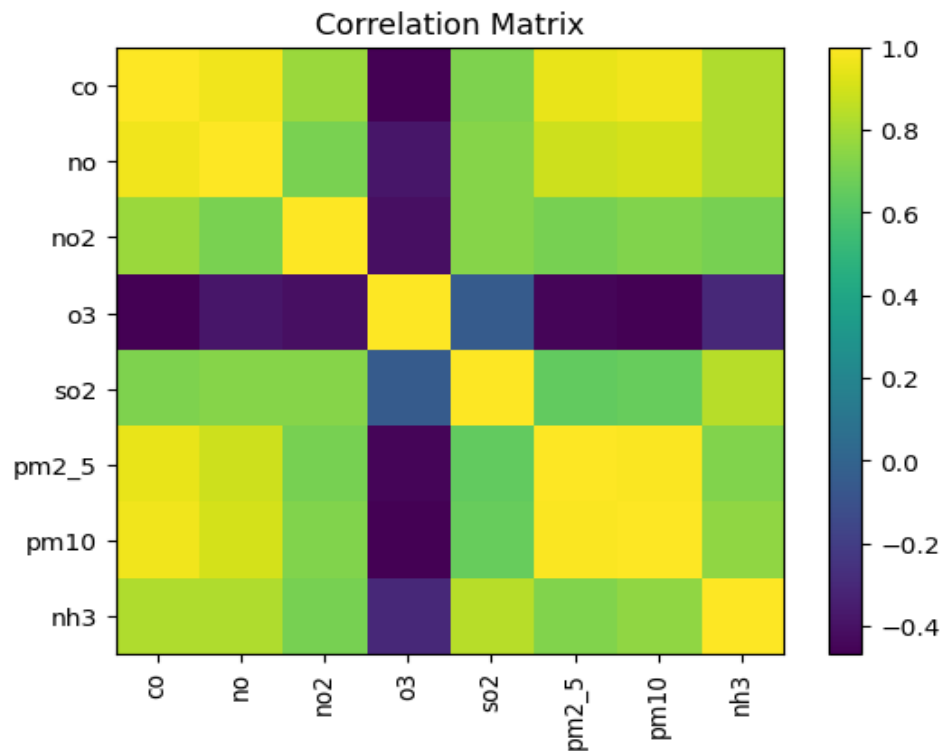
```
hourly_avg = df.groupby('hour')['pm2_5'].mean()  
hourly_avg.plot()
```



Pollution peaks during evening and early morning hours, likely due to vehicular emissions and reduced atmospheric dispersion.

6. Correlation Analysis

```
corr = df[['co','no','no2','o3','so2','pm2_5','pm10','nh3']].corr()
```



Strong positive correlation between PM2.5 and PM10 indicates particulate dominance. CO and NO2 correlations suggest combustion-related pollution sources.